

IoT-Based Smart Home Automation System Using ESP32

Anil Kumar Senapati

June 2, 2026

Skillbit Technologies Internship Project Report

Contents

1	Abstract	3
2	Introduction	3
3	Objectives	3
4	Components Required	3
5	Circuit	4
6	Methodology	4
7	Working Principle	4
8	Software Implementation	5
8.1	Pin Configuration	5
9	Results	5
10	Advantages	5
11	Applications	6
12	Future Scope	6
13	Conclusion	6
14	References	6

1 Abstract

The Internet of Things (IoT) has revolutionized the way household appliances are monitored and controlled. This project presents an IoT-Based Smart Home Automation System using the ESP32 microcontroller and a 2-channel relay module. The system enables users to remotely control electrical appliances such as lights and fans through a web browser over a Wi-Fi network. The ESP32 hosts a local web server that provides an interactive interface for appliance control. The proposed system is low-cost, energy-efficient, scalable, and suitable for modern smart home applications.

2 Introduction

Home automation refers to the automatic and intelligent control of household devices and appliances. With the rapid advancement of IoT technology, it has become possible to monitor and control home appliances remotely using smartphones, computers, and web-based interfaces.

The objective of this project is to design and implement a smart home automation system that allows users to switch appliances ON and OFF remotely through a Wi-Fi-enabled ESP32 microcontroller. The system improves convenience, reduces manual effort, and serves as a foundation for advanced smart home technologies.

3 Objectives

- To develop a smart home automation system using ESP32.
- To remotely control electrical appliances using a web interface.
- To demonstrate IoT-based communication over Wi-Fi.
- To provide a low-cost and scalable automation solution.
- To improve user convenience and energy management.

4 Components Required

Sl. No.	Component	Quantity
1	ESP32 Development Board	1
2	2-Channel Relay Module	1
3	AC Bulb	1
4	AC Fan	1
5	Jumper Wires	As Required
6	Wi-Fi Network	1
7	Power Supply	1

5 Circuit

Figure 1 shows the circuit diagram of the proposed IoT-based Smart Home Automation System.

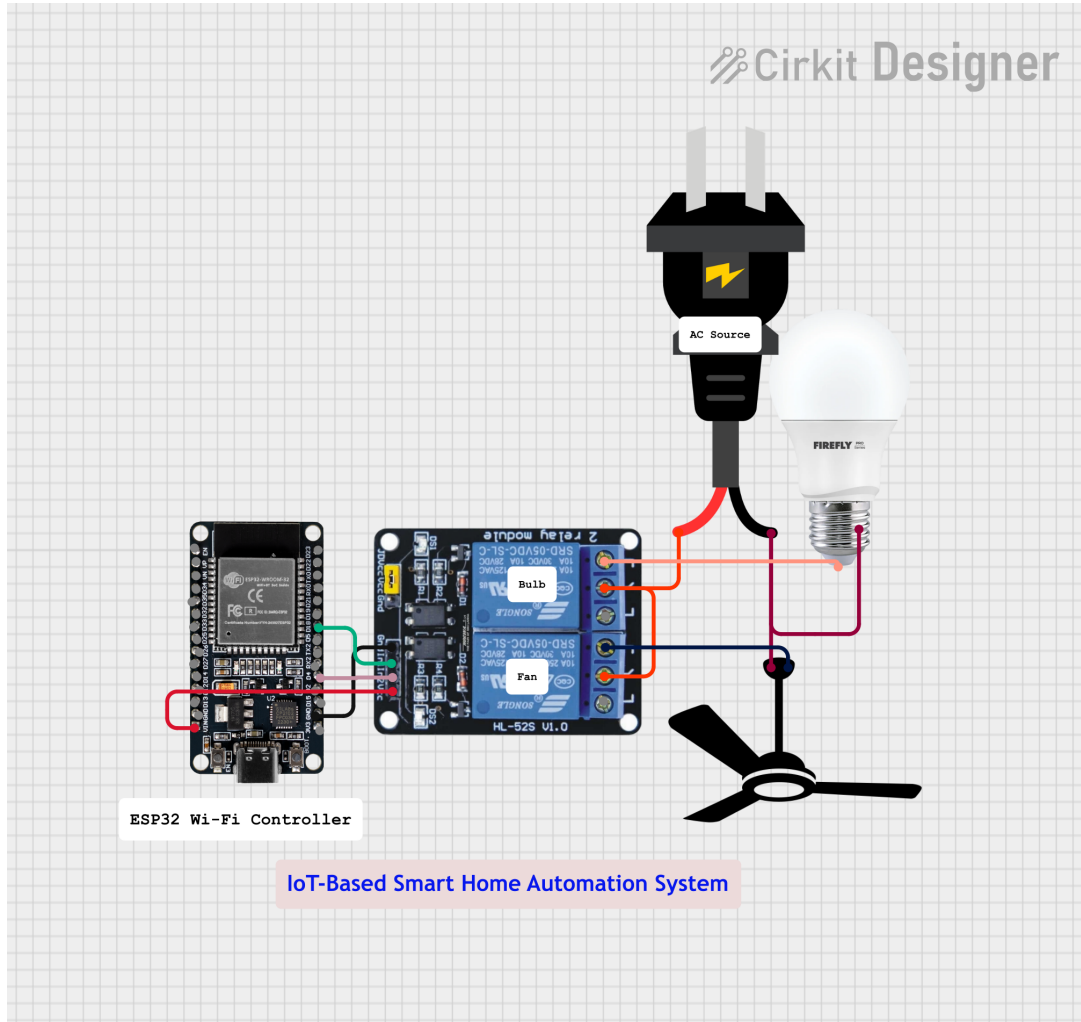


Figure 1: Circuit Diagram of IoT-Based Smart Home Automation System

6 Methodology

The project is implemented using the ESP32 microcontroller and a relay module. The ESP32 connects to a local Wi-Fi network and hosts a web server. Users access the web interface through the ESP32 IP address using any web browser.

The relay module acts as a switching device that controls the connected appliances. When the user presses an ON/OFF button on the web interface, the ESP32 processes the request and activates or deactivates the corresponding relay channel.

7 Working Principle

The system operates according to the following steps:

1. ESP32 connects to the Wi-Fi network.
2. A local web server is started on the ESP32.
3. The user enters the ESP32 IP address in a web browser.
4. A control interface containing ON/OFF buttons is displayed.
5. Commands sent by the user are received by the ESP32.
6. Relay channels switch the connected bulb and fan accordingly.
7. Appliance status changes are reflected immediately.

8 Software Implementation

The project is programmed using Arduino IDE. The ESP32 web server handles incoming requests and controls the relay outputs connected to GPIO18 and GPIO4.

8.1 Pin Configuration

Appliance	GPIO Pin
Bulb Relay	GPIO18
Fan Relay	GPIO4

9 Results

The proposed smart home automation system was successfully designed and implemented. The ESP32 established communication over Wi-Fi and hosted a web-based control interface.

The connected appliances (bulb and fan) were controlled remotely through a web browser. The relay module responded correctly to user commands, demonstrating reliable switching functionality. The system achieved real-time control with minimal latency and stable operation.

10 Advantages

- Remote appliance control.
- Low-cost implementation.
- Easy to install and maintain.
- Energy-efficient operation.
- Scalable architecture.
- Supports future smart home upgrades.

11 Applications

- Smart homes
- Home energy management
- Remote appliance monitoring
- Elderly assistance systems
- Office automation
- Hostel and apartment automation

12 Future Scope

- Google Assistant integration.
- Mobile application development.
- Energy consumption monitoring.
- AI-based appliance scheduling.
- Cloud-based monitoring and control.
- Voice-controlled home automation.

13 Conclusion

This project successfully demonstrates an IoT-Based Smart Home Automation System using ESP32 and a relay module. The developed system enables users to remotely control household appliances through a web browser using Wi-Fi communication. The implementation is cost-effective, reliable, and suitable for real-world smart home environments. The project also provides a strong foundation for future enhancements such as voice assistants, cloud connectivity, and intelligent automation.

14 References

1. ESP32 Technical Reference Manual.
2. Arduino IDE Documentation.
3. ESP32 Wi-Fi Library Documentation.
4. Internet of Things (IoT) Fundamentals.
5. Relay Module User Guide.